IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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04/03/2026

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Data collection
- Data wrangling
- Exploratory Data Analysis with Data Visualization
- Exploratory Data Analysis with SQL
- Building an interactive map with Folium
- Building a Dashboard with Plotly Dash
  Predictive analysis (Classification)
- Summary of all results
- Exploratory Data Analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Introduction

- SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upward of 165 million dollars each, much of the savings is because SpaceX can reuse the first stage.
- Therefore if we can determine if the first stage will land, we can determine the cost of a launch.
- This information can be used if an alternate company wants to bid against SpaceX for a rocket launch.
- We will predict if the Falcon 9 first stage will land successfully.

Section 1

Methodology

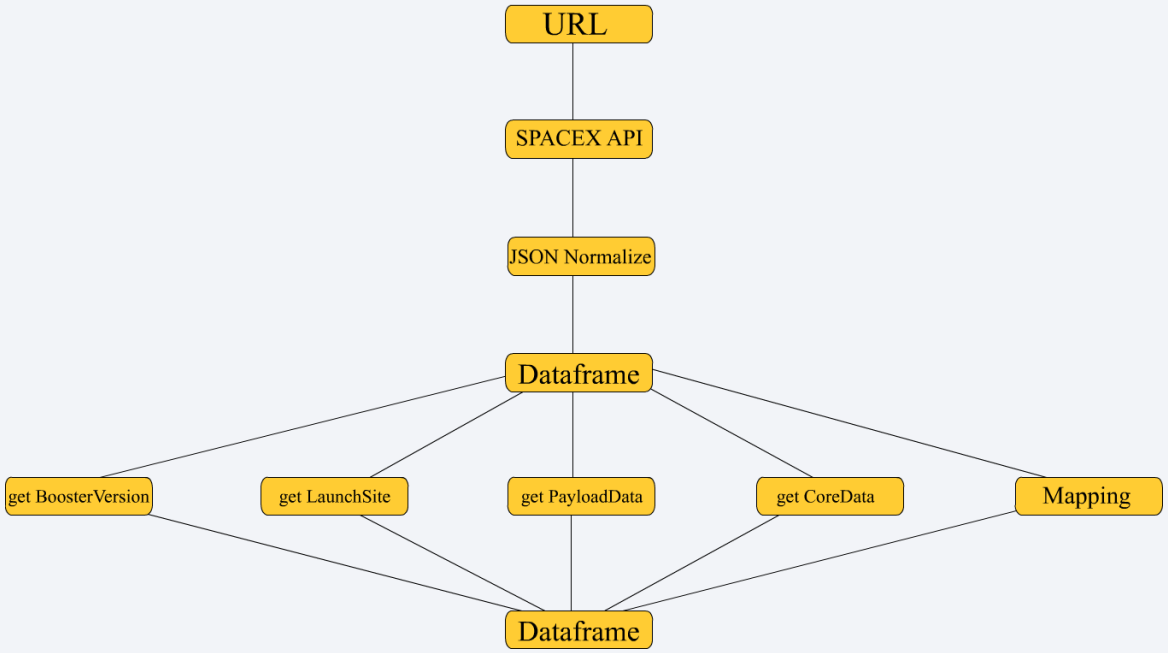
Methodology

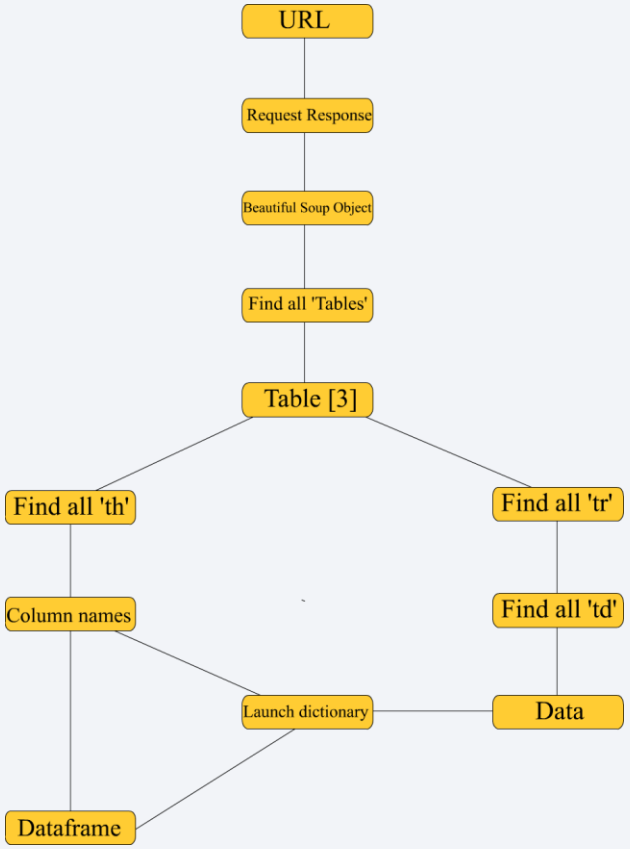
Executive Summary

- Data collection methodology:
  - Using SpaceX Rest API
  - Using Web Scrapping from Wikipedia Performed
- Perform data wrangling
  - Filtering the data
  - Dealing with missing values
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Predictive analysis using a classification model, hyperparameter tuning and evaluating the best results.

Data Collection

- The data was collected using an API from SPACEX and the data was also webscraped from the SPACEX website.

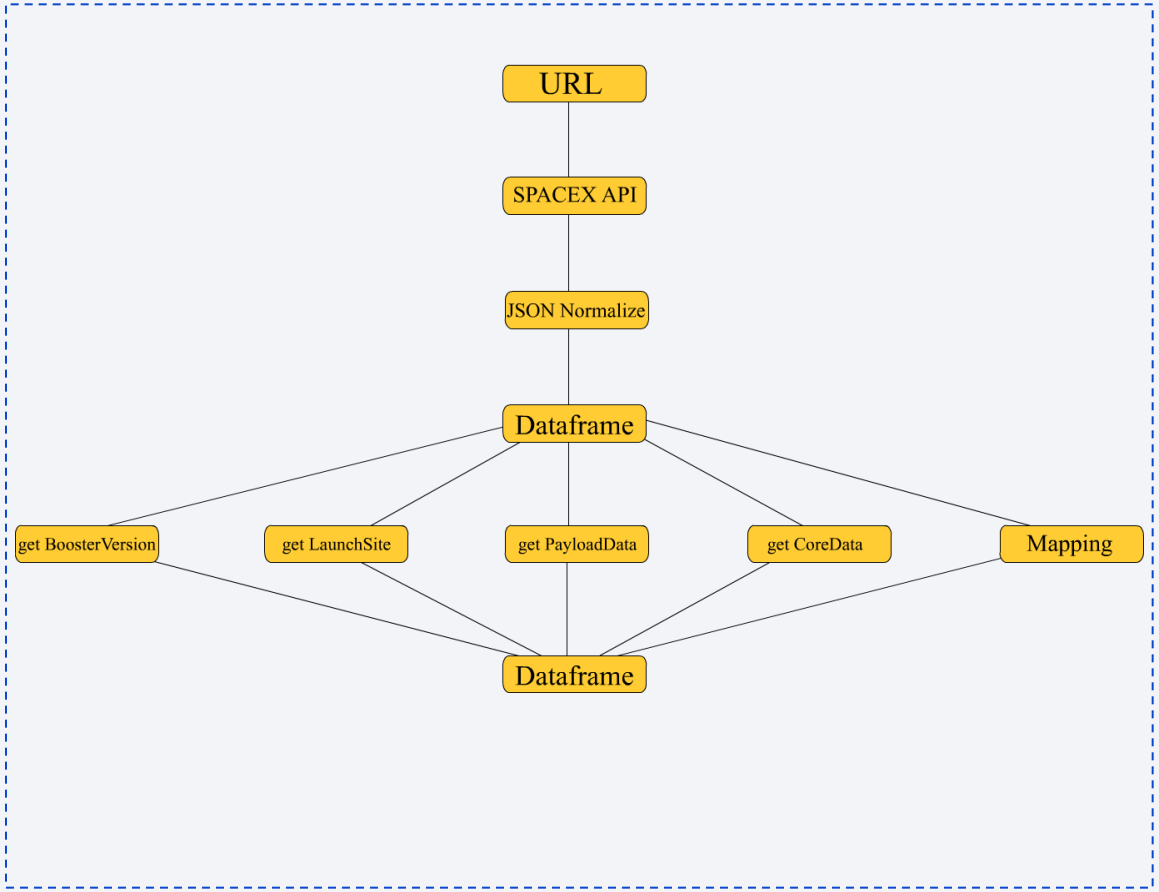




Data Collection – SpaceX API

- A URL was used in a REST API call to SPACEX and the response was converted from JSON into a dataframe. Functions were applied to the dataframe to be able to get specific column data from that dataframe and a new dataframe was formed from those columns.

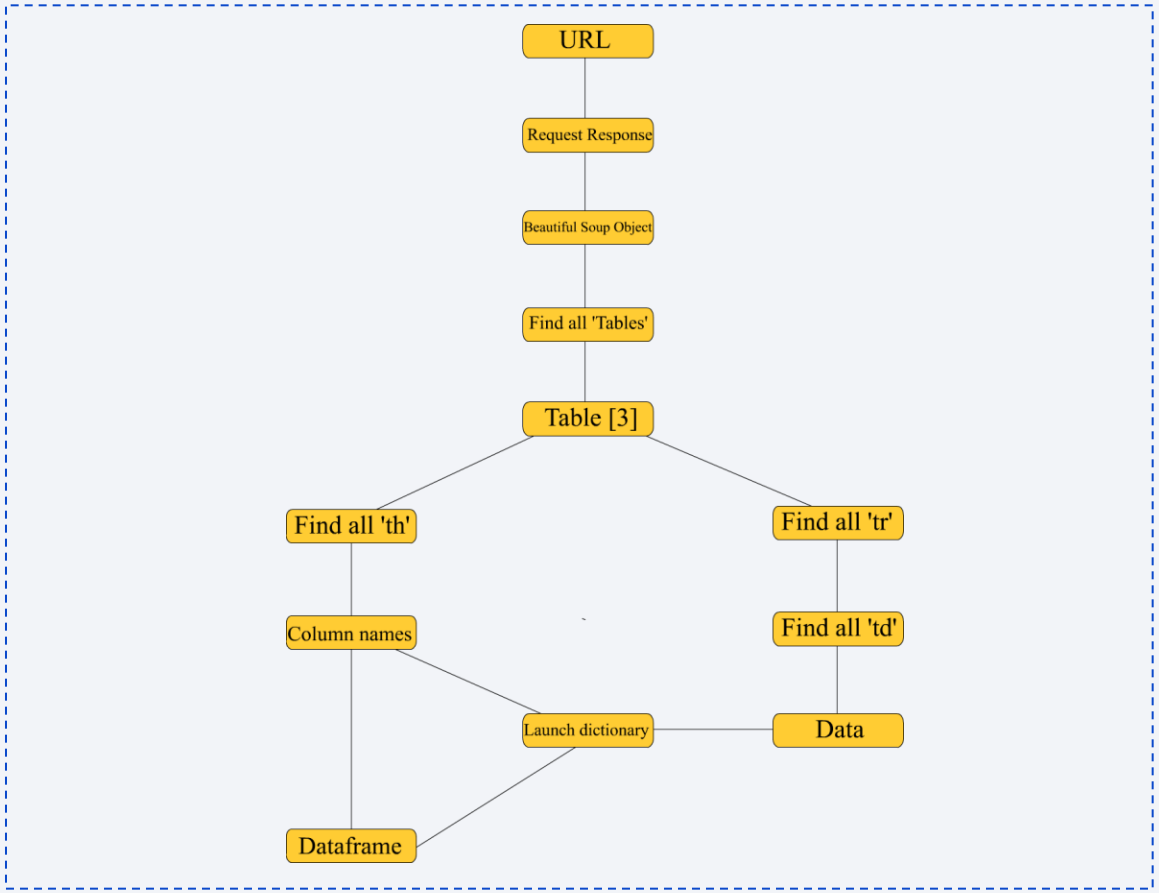
- https://github.com/PrnceTheKiiD/Winning-Space-Race-with-Data-Science/blob/main/Notebooks/jupyter-labs-spacex-data-collection-api-v2.ipynb



Data Collection - Scraping

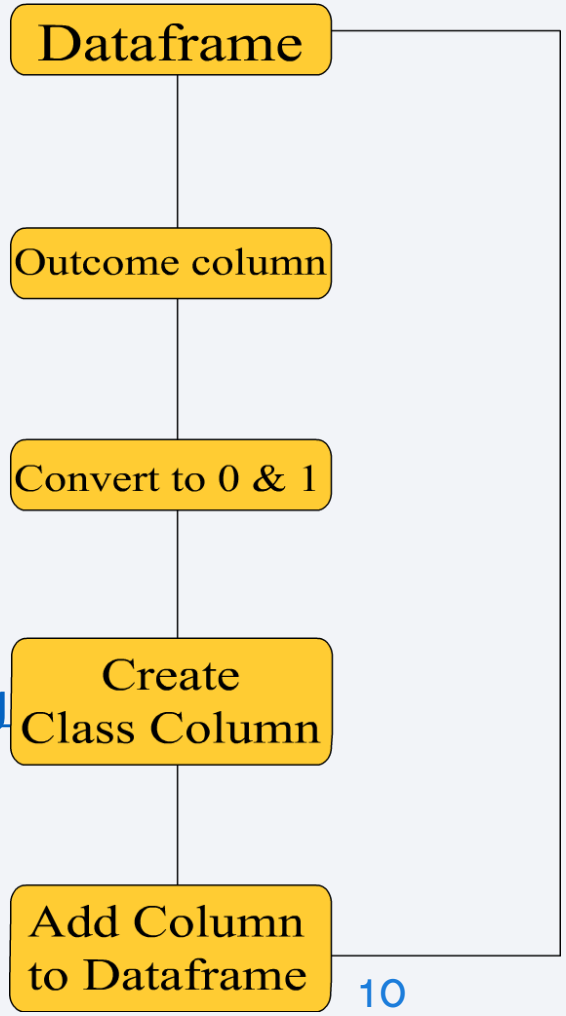
- A URL was used to get an response which was turned into a Beautiful Soup object where all tables were found and the 4th table was isolated to get data from. All 'th's were found in the table and a column names were extracted to make a new dataframe. All 'td' in all 'tr's was passed into a dictionary and a new dataframe was created from the column names and dictionary.

- https://github.com/PrnceTheKiiD/Winning-Space-Race-with-Data-Science/blob/main/Notebooks/jupyter-labs-webscraping.ipynb



Data Wrangling

- A dataframe with the launch sites was used and the Outcome column is extracted.
- The Outcome column was then converted into a column of 0 and 1 values based off of the landing outcome.
- The converted column was stored as the class column
- The class column was added to the original dataframe with launch sites on it
- https://github.com/PrnceTheKiiD/Winning-Space-Race-with-Data-Science/blob/main/Notebooks/labs-jupyter-spacex-Data%20wrangling v2.ipynb



EDA with Data Visualization

- The relationship between Flight Number and Launch Site was visualized in a scatter plot to clearly see the distribution of flight numbers amongst sites.
- The relationship between Payload and Launch Site was visualized in a scatter plot to clearly see the distribution of payloads amongst sites.
- The relationship between success rate of each orbit type was visualized in a bar plot to be able to compare the success rate of each orbit type.
- The relationship between FlightNumber and Orbit type was visualized in a scatter plot to clearly see the distribution of flight numbers according to the orbit type.
- Visualize the relationship between Payload and Orbit type was visualized in a scatter plot to clearly see the distribution of payloads according to the orbit type.
- The launch success yearly trend was visualized in a line plot to be able to see the changes in trend of the success rate.
- https://github.com/PrnceTheKiiD/Winning-Space-Race-with-Data-Science/blob/main/Notebooks/jupyter-labs-eda-dataviz-v2.ipynb

EDA with SQL

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA'
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date when the first successful landing outcome in ground pad was achieved.
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List all the booster versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.
- List the records which will display the month names, failure landingoutcomes in drone ship ,booster versions, launchsite for the months in year 2015.
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.
- https://github.com/PrnceTheKiiD/Winning-Space-Race-with-Data-Science/blob/main/Notebooks/jupyter-labs-eda-sql-coursera_sqllite.ipynb

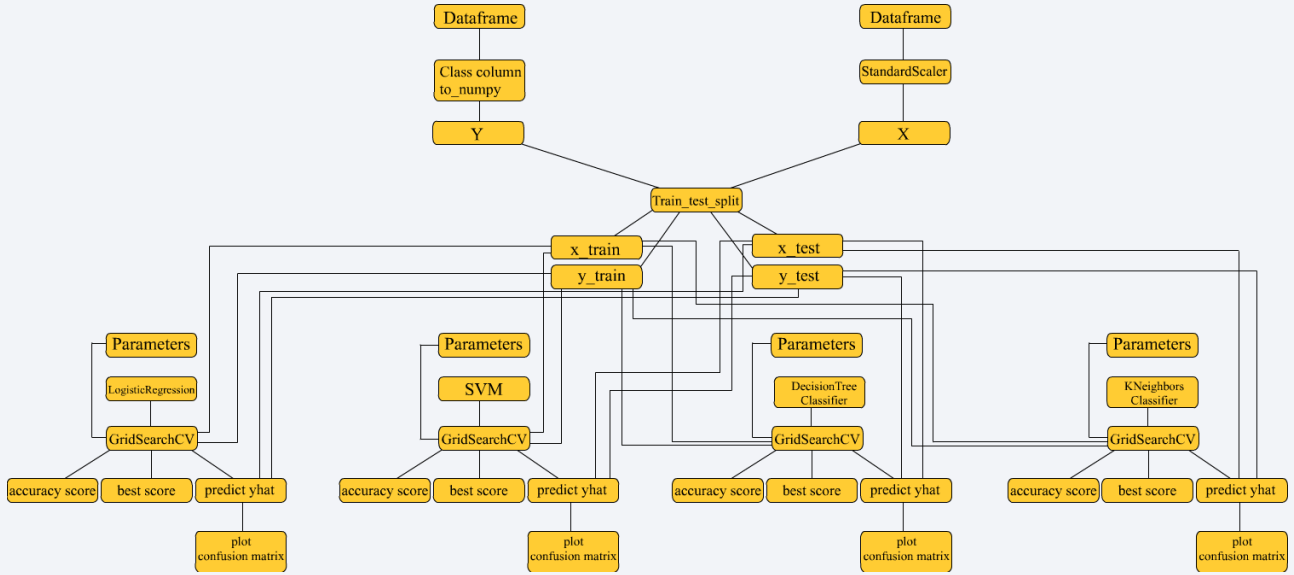
Build an Interactive Map with Folium

- These are the objects that were added to the Map: Marker, Marker Cluster, Circle, Polyline, Icon, DivIcon and MousePosition.
- The objects were added to the Map to be able to identify the launch sites, the the successful and failed launches of the site on the map and to see the details on the site's proximity to significant landmarks.
- https://github.com/PrnceTheKiiD/Winning-Space-Race-with-Data-Science/blob/main/Notebooks/lab-jupyter-launch-site-location-v2.ipynb

Build a Dashboard with Plotly Dash

- A Pie chart that shows the total successful launches of all sites.
- The Pie chart is connected to a dropdown that allows us to select the launch site
- A Scatter plot that shows the landing outcome of each booster version according to the payload
- The Scatterplot is connected to the launch site dropdown allowing us to select the launch site and the Scatterplot is also connected to a RangeSlider that allows us to see the effect of payload mass on the success rate of each booster version
- https://github.com/PrnceTheKiiD/Winning-Space-Race-with-Data-Science/blob/main/Notebooks/spacex_dash_app.py

Predictive Analysis (Classification)



- The launch site dataset had the class column extracted and converted to_numpy and the array was stored in Y. The feature engineered dataframe with OneHotEncoding was scaled using StandardScaler and stored as X. The X and Y dataset was split using train_test_split into training and testing datasets. The training data was used in a GridSearchCV for a LogisticRegession, SVM, DecisionTree Classifier and KNeighbors Classifier algorithm to check the parameters given. The best accuracy score was attained and the yhat is predicted and was stored in the yhat. The yhat and test datasets are used to plot the confusion matrix.

- https://github.com/PrnceTheKiiD/Winning-Space-Race-with-Data-Science/blob/main/Notebooks/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

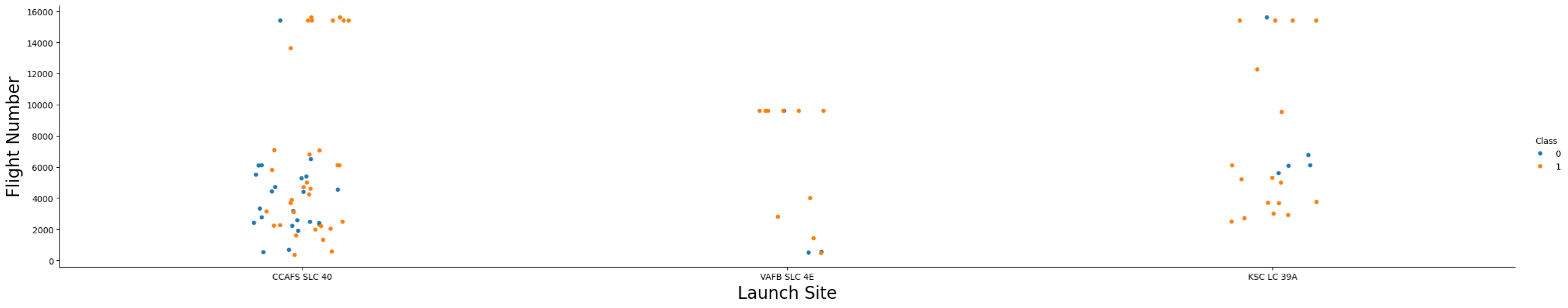
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

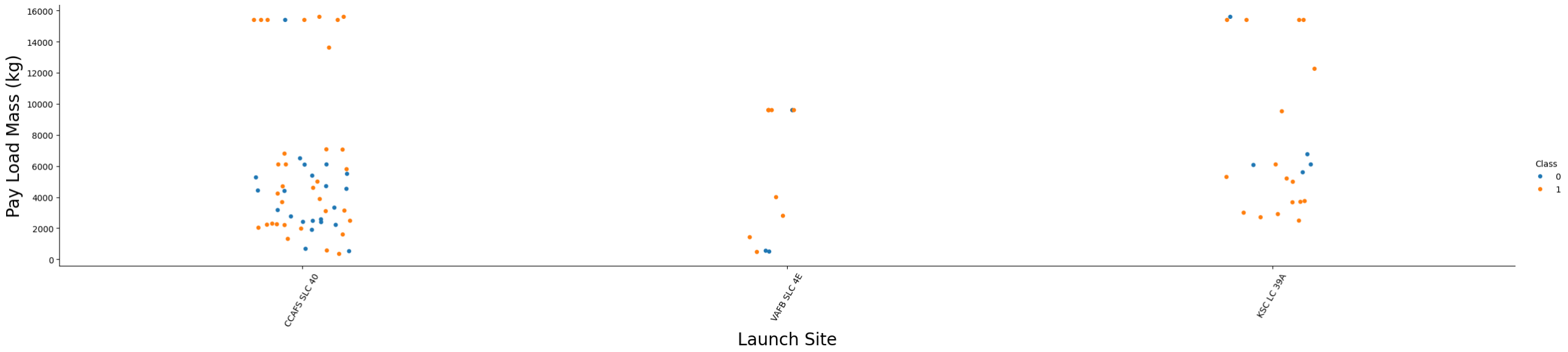
Insights drawn from EDA

Flight Number vs. Launch Site



- The CCAFS SLC 40 launch site has had more success with the flight numbers around 15000
- VAFB SLC 4E has more successful landings with flight numbers around 10000
- KSC LC 39A more success with flight numbers around 15000

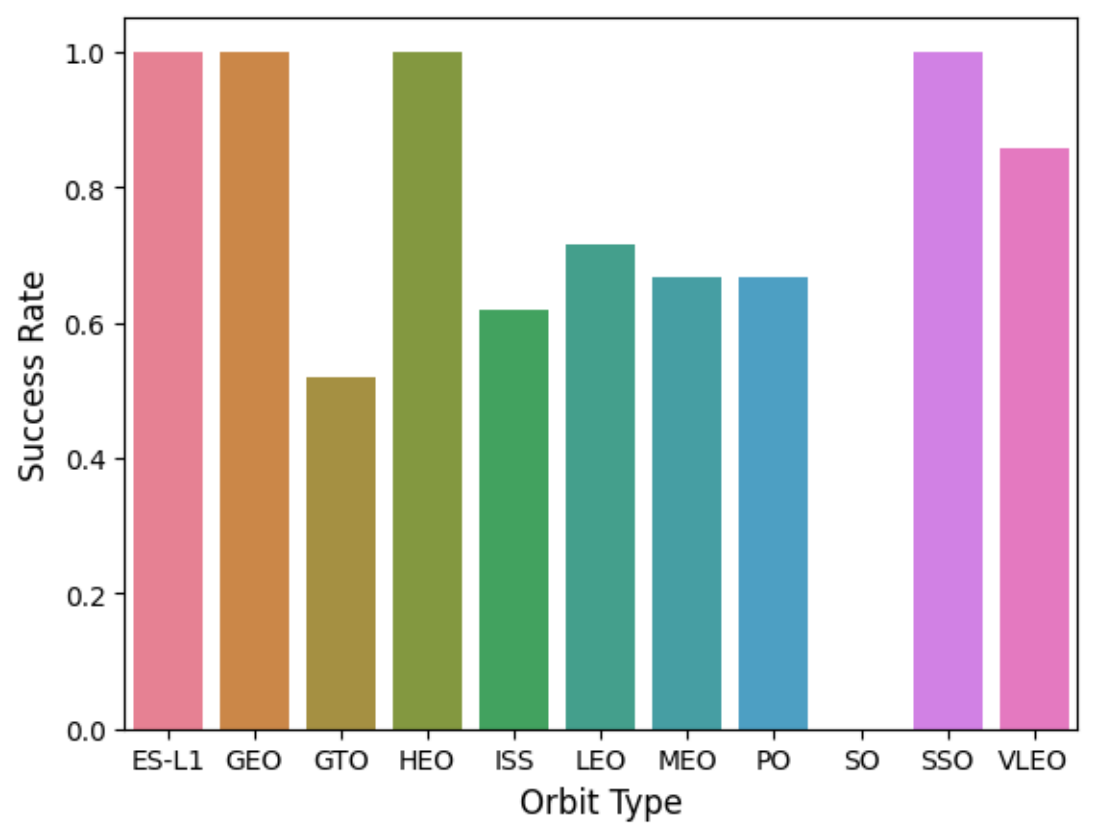
Payload vs. Launch Site



- The CCAFS SLC 40 launch site has had more success with the payloads around 15000 kg.
- VAFB SLC 4E has more successful landings with payloads around 10000 kg.
- KSC LC 39A more success with payloads around 15000 kg.

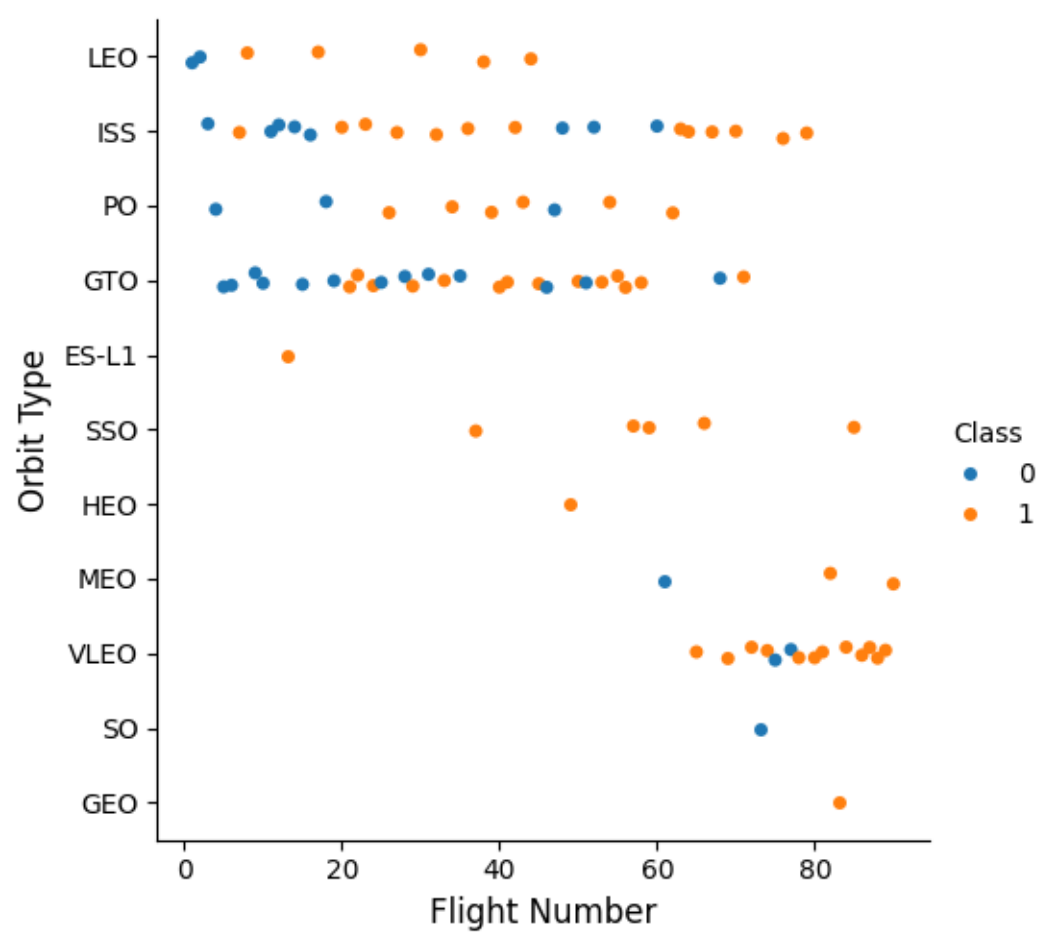
Success Rate vs. Orbit Type

- The orbit types that have the highest success rate are, ES-L1, GEO, HEO and SSO.
- The lowest success rate belongs to the SO orbit type.
- The orbit types GTO, ISS, LEO, MEO, PO and VLEO have an average success rate around 0.5 to 0.8



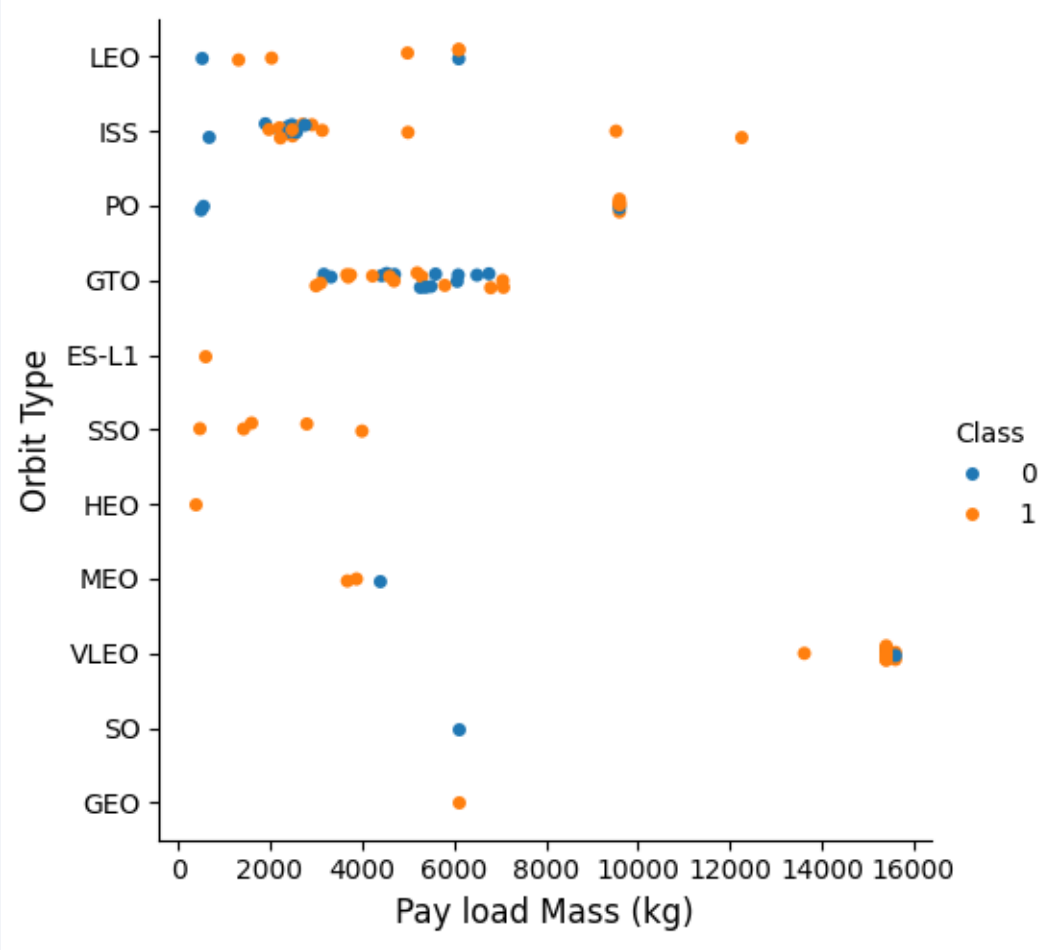
Flight Number vs. Orbit Type

- ISS has the most success with flight numbers from 20 to 40 and 60 to 80.
- GTO has the most success from flight number 20 to 60
- VLEO has the most success from flight number 60 to 90
- PO has the most success from flight number 20 to 70
- LEO has the most success from flight number 10 to 50
- SSO has had success in all flight numbers from 30



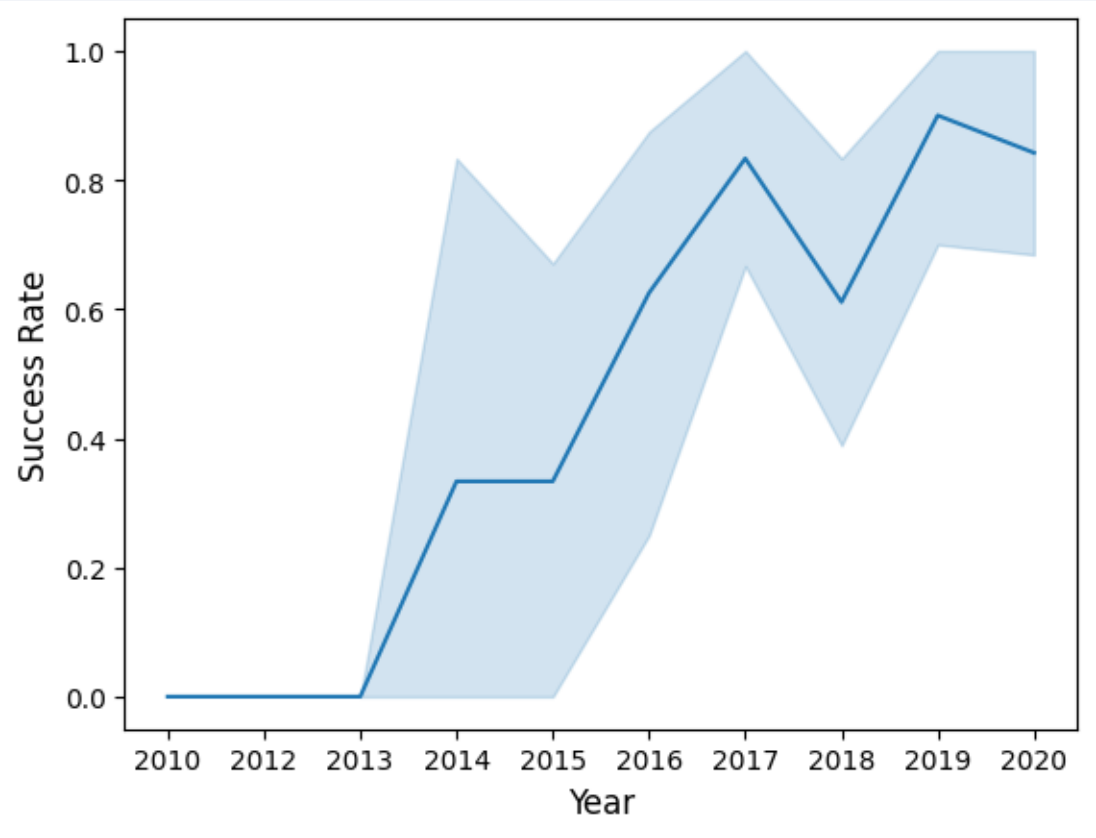
Payload vs. Orbit Type

- LEO has more success with payloads from 2000kg to 6000kg
- ISS has more success with payloads from 2000kg to 12000kg
- PO only has success with a payload of 10000kg
- SSO has success in all launches from 3000kg to 8000kg
- MEO mainly has success around 3600kg to 4000kg
- VLEO has success from 13700kg to 15700kg



Launch Success Yearly Trend

- The success rate has been on an upward trend from 2010 to 2019 with the exception of 2018 dropping in success rate from 0.8 in 2017 to 0.6 in 2018 which then picked back up to again in 2019



All Launch Site Names

```
[43]
%sql SELECT DISTINCT Launch_Site from SPACEXTBL
```

Python

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- All of the launch sites that were analyzed

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5
```

[44] Python

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- A list of the 5 records where launch sites begin with `CCA`

Total Payload Mass

```
[40]  %sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE CUSTOMER LIKE 'NASA (CRS)%'
                                                                                        Python
...  * sqlite:///my_data1.db
     Done.
...
```

SUM(PAYLOAD_MASS__KG_)
48213

- The total payload mass for the customer NASA (CRS) is 48213 kg

Average Payload Mass by F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 v1.1%'
```

Python

```
 * sqlite:///my_data1.db
Done.
```

AVG(PAYLOAD_MASS__KG_)
2534.6666666666665

- The average payload mass carried by booster version F9 v1.1 is 234,6666666666665kg

First Successful Ground Landing Date

```
[47]
%sql SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome LIKE 'Success (ground pad)'
                                                                                    Python
```

```
 * sqlite:///my_data1.db
Done.
```

MIN(Date)
2015-12-22

- The first date on which a landing was successful on a ground pad

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql SELECT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome LIKE 'Success (drone pad)' AND PAYLOAD_MASS__KG_ >4000 OR PAYLOAD_MASS__KG_ <6000
```

[15] ✓ 0.0s Python

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 v1.0 B0003
F9 v1.0 B0004
F9 v1.0 B0005
F9 v1.0 B0006
F9 v1.0 B0007
F9 v1.1 B1003
F9 v1.1
F9 v1.1
F9 v1.1
F9 v1.1
F9 v1.1
F9 v1.1 B1011
F9 v1.1 B1010
F9 v1.1 B1012
F9 v1.1 B1013
F9 v1.1 B1014
F9 v1.1 B1015
F9 v1.1 B1016
F9 v1.1 B1018
F9 FT B1019

- A list of boosters which have successfully landed on a drone ship and had payload mass greater than 4000 but less than 6000

Total Number of Successful and Failure Mission Outcomes

```
%sql SELECT Mission_Outcome, COUNT(*) AS Total FROM SPACEXTBL GROUP BY Mission_Outcome
```

[16] ✓ 0.0s Python

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

- The total number value count of successful and failure mission outcomes

Boosters Carried Maximum Payload

```
%sql SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ =  (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL)
```

[17] ✓ 0.0s Python

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- A list of the boosters which have carried the maximum payload mass

2015 Launch Records

```
%sql SELECT substr(Date, 6, 2) as month, Landing_Outcome, Booster_Version, Launch_Site from SPACEXTABLE where substr(Date, 0, 5) = '2015' and Landing_Outcome = 'Failure (drone ship)
```

[18] ✓ 0.0s Python

```
 * sqlite:///my_data1.db
Done.
```

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- A list of the failed landing outcomes in a drone ship, their booster versions, and launch site names for in year 2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql select Landing_Outcome, COUNT(*) as Count from SPACEXTBL where Date between '2010-06-04' and '2017-03-20' group by Landing_Outcome order by Count Desc
```

[19] ✓ 0.0s Python

```
 * sqlite:///my_data1.db
Done.
```

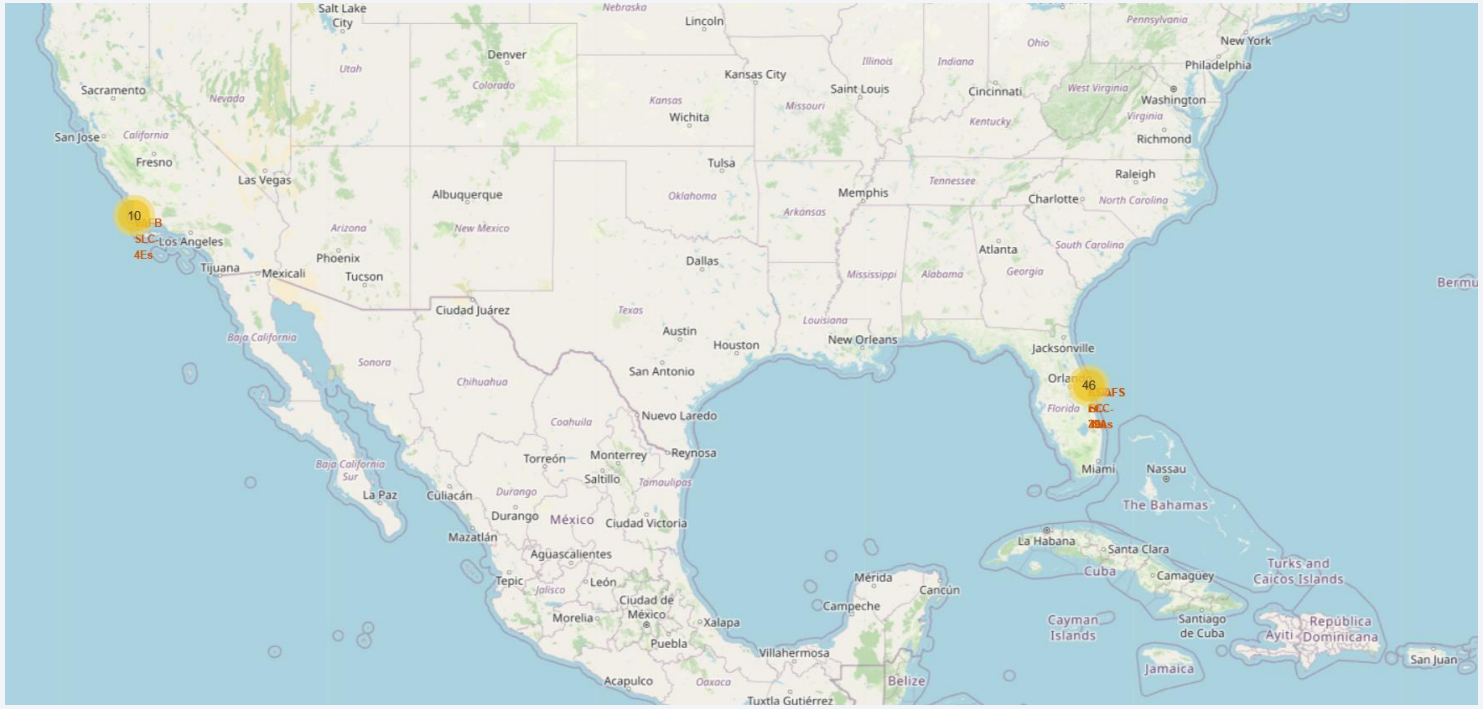
Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

- A list of each landing outcome's count between the date 2010-06-04 and 2017-03-20 that is ranked in descending order

Section 3

Launch Sites Proximities Analysis

Launch Sites Globally

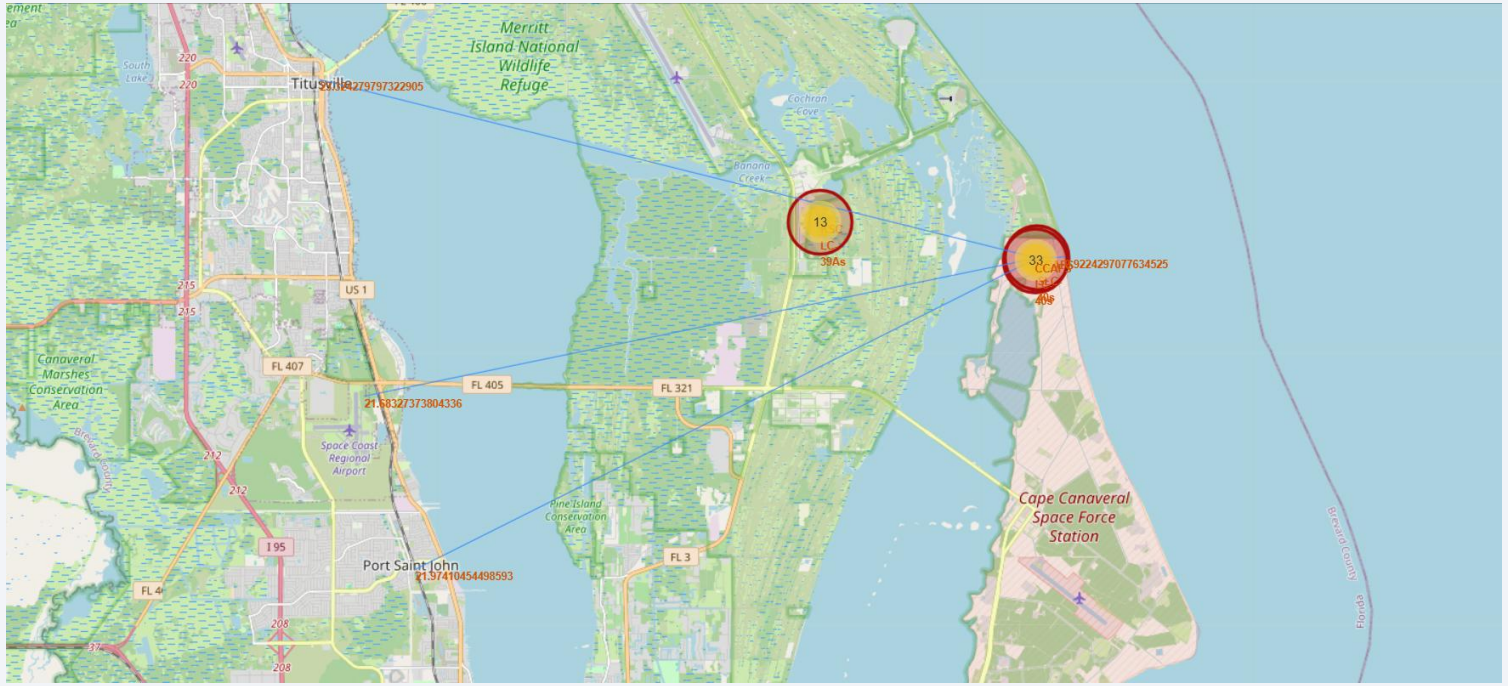


- All launch sites' location markers on a global map
- CCAFS LC-40 and CCAFS SLC-40 are near the coastline of False Bay
- KSC LC-39A is near Banan Creek
- VAFB SLC-4E is near Vandenburg State Marine Reserve

Launch Site Outcomes

- Color-labeled launch outcomes on the map
- The green markers icons represent a success and the red markers represent a failure.
- The launch outcomes of CCAFS SLC-40 are displayed which is the successful launch site

Launch Site Proximity



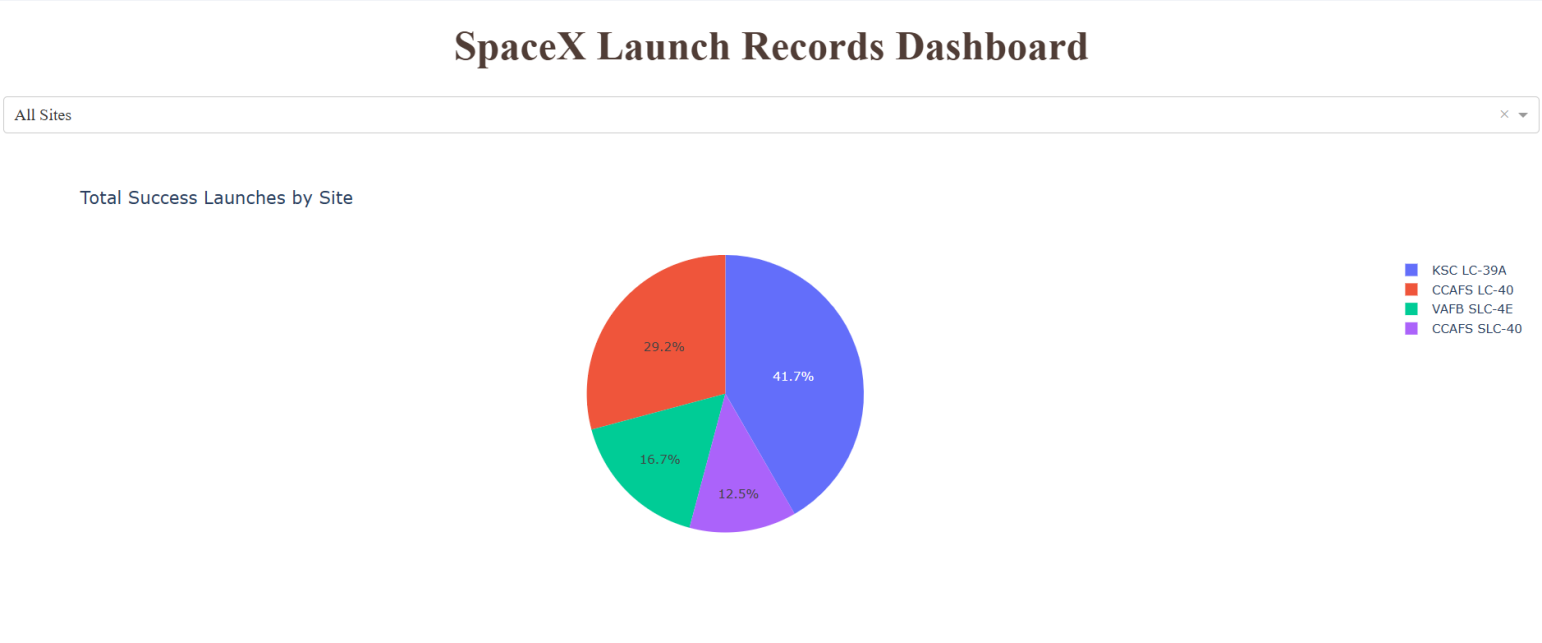
- CCAFS SLC-40 is 0.922429707763452 5km away in proximity to a coastline
- CCAFS SLC-40 is 23.324279797322905km away in proximity to a close city
- CCAFS SLC-40 is 21.68327373804336km away in proximity to a highway
- CCAFS SLC-40 21.97410454498593km is away in proximity to a railway

Section 4

Build a Dashboard with Plotly Dash

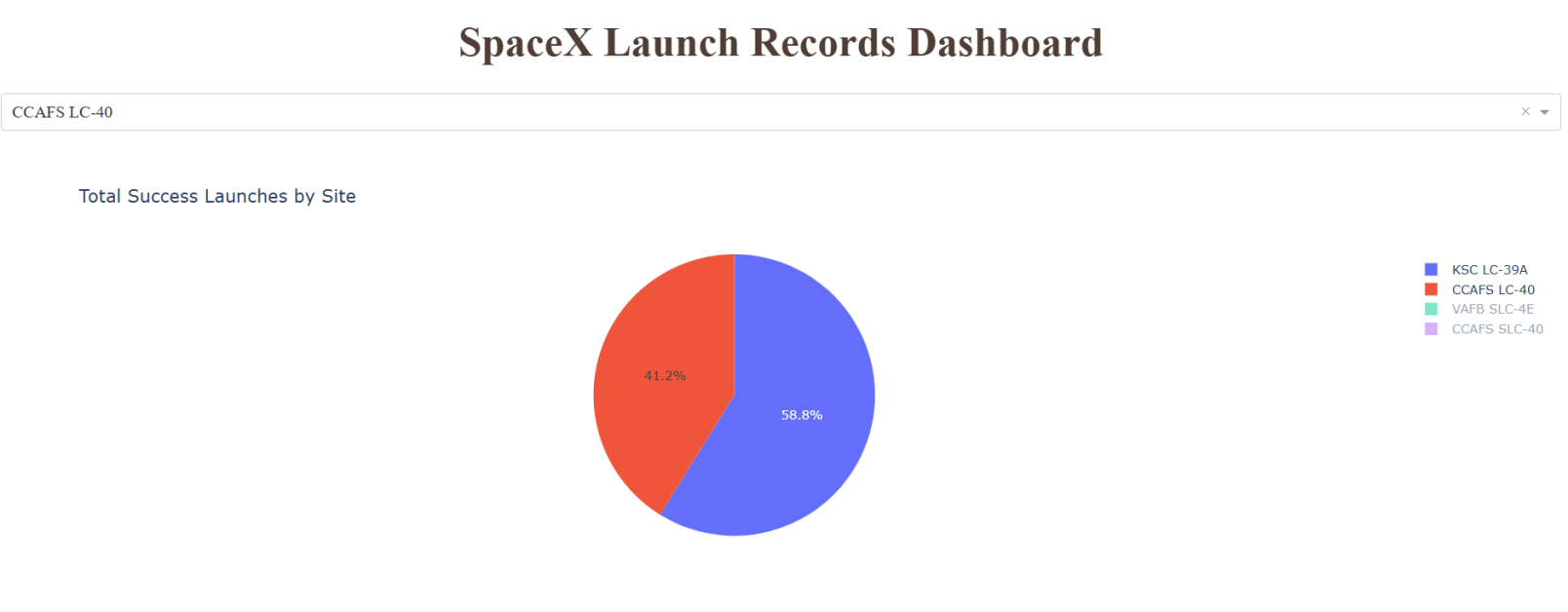
Total Success Launches by Site

- KSC LC 39A has the most success with 10 launches which is 41.7% of the launches
- CCAFS LC-40 has 7 successful launches which is 29.2% of the launches
- VAFB SLC-4E has 4 successful launches which is 16.7% of the launches
- CCAFS SLC-40 has the least success with 3 launches which is 41.7% of the launches

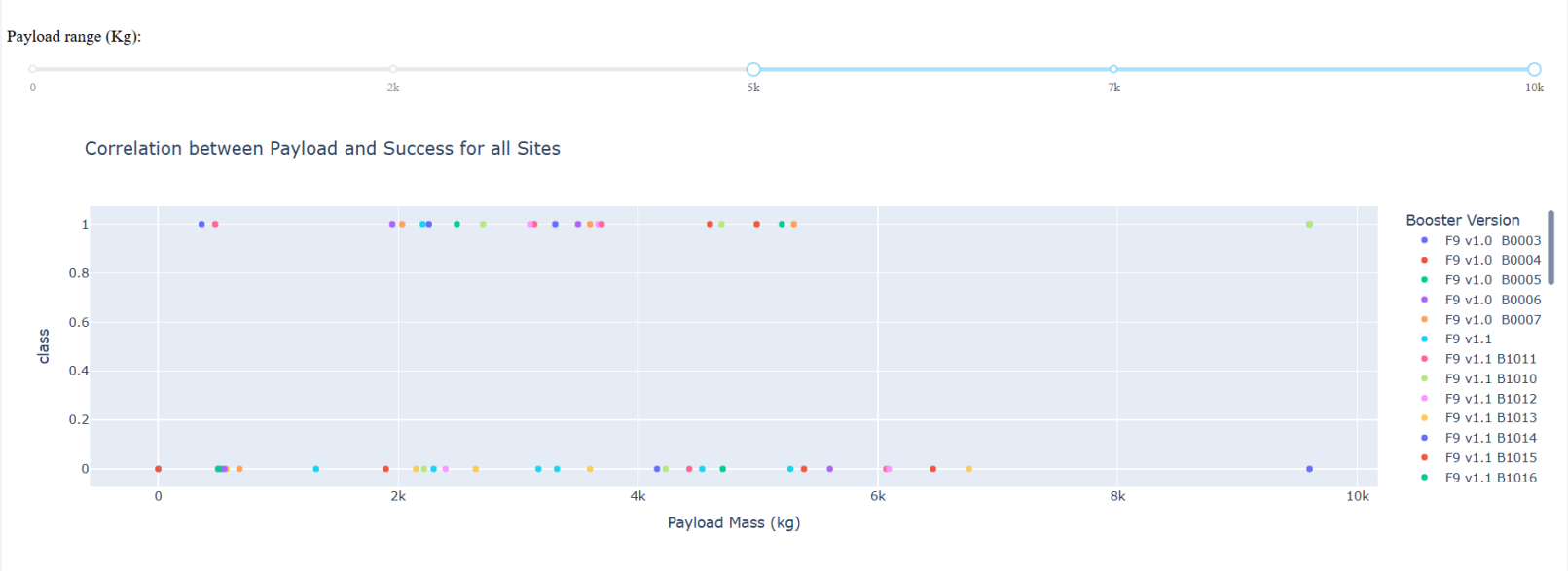


Total Success Launches of launch site KSC LC-39A

- KSC LC 39A has the highest success ratio



Correlation between Payload and Success for all Sites



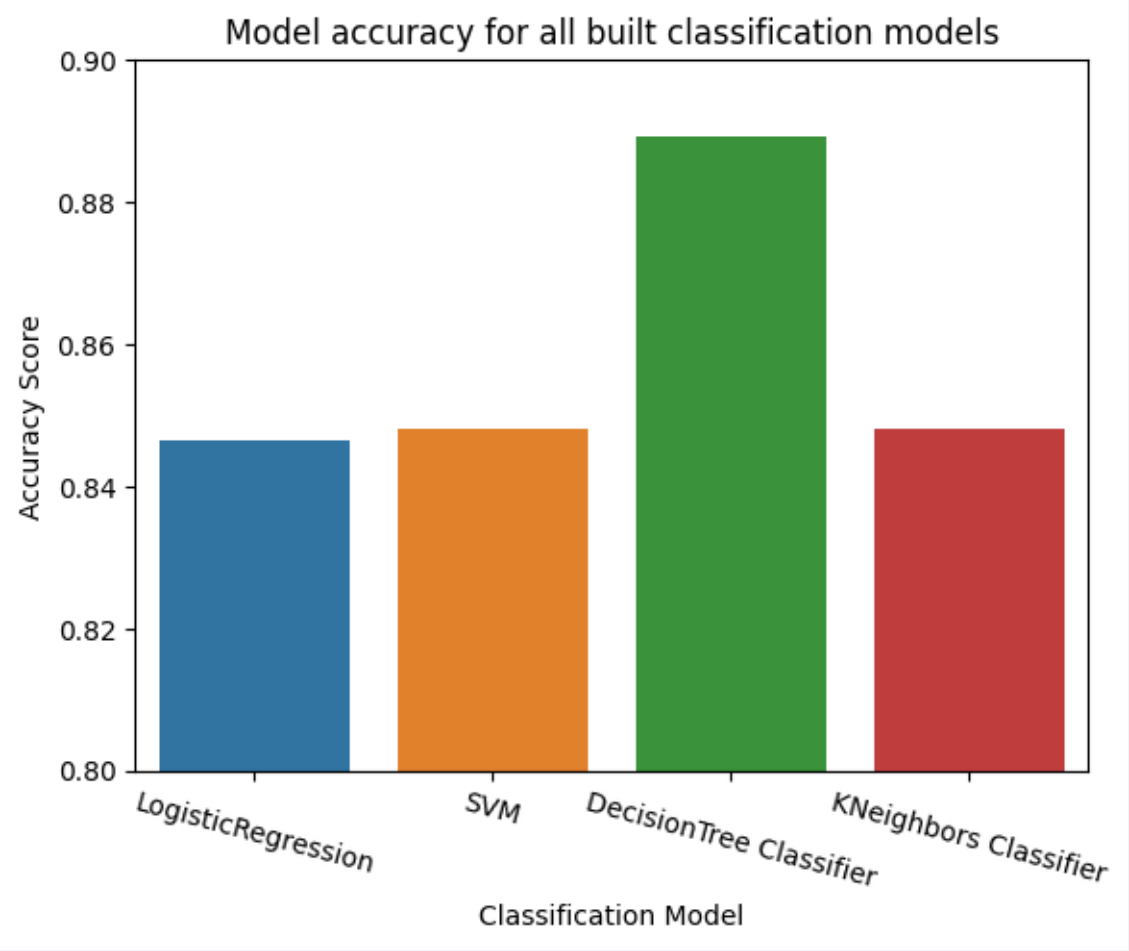
- The boosters with the highest success having large payloads are F9 FT B1026 with 4600kg, F9 FT B1022 with 4696kg, F9 B4 B1040 with 4990kg, F9 FT B1031.2 with 5300kg and F9 B4 B1041.1 with 9600kg as the highest
- The boosters with the highest success having low payloads are F9 FT B1038.2 with 475kg and F9 B4 B10465.1 with 362kg as the lowest
- The majority of the successful boosters ranged from a payload of 362kg to 5300kg with the exception of booster F9 B4 B1041.1 being the biggest payload having 9600kg

Section 5

Predictive Analysis (Classification)

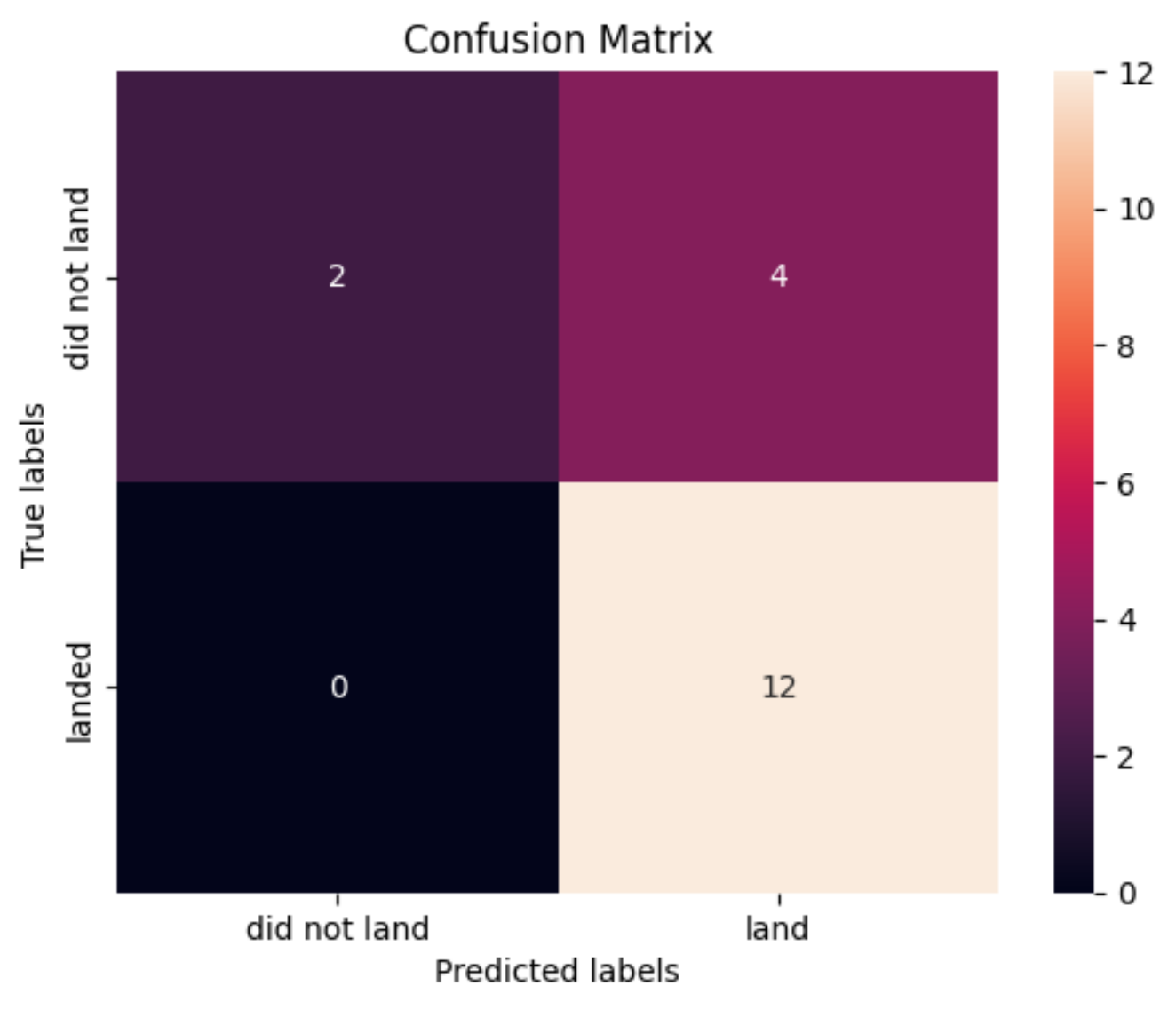
Classification Accuracy

- The Decision Tree Classifier has the highest accuracy score amongst all the models with an accuracy score of 0.862500
- The other models range around an accuracy score of 0.846429 to 0.848214



Confusion Matrix

- The Decision Tree Classifier had:
- 12 True Positives
- 0 False Positives
- 2 False Negatives
- 4 False Positives



Conclusions

- The Falcon 9 will have a successful launch and landing.
- The Falcon 9 will be most successful with payloads that are a maximum of 9600kg and lower
- The Falcon 9 will be most successful with the booster F9 B4 B1041.1 to be able to handle a. higher payloads
- The optimal launch site for the Falcon 9 is the CCAFS SLC 40 launch site.

Appendix

- dataset_part_1 (Dataset)
- dataset_part_2 (Dataset)
- dataset_part_3 (Dataset)
- spacex_launch_geo (Dataset)
- spacex_launch_dash (Dataset)
- my_data1 (Database)
- Success Rate of Orbit Type (Barplot)
- Success Rate Yearly (Lineplot)
- Payload Mass by Orbit Type (Scatterplot)
- Flight Number by Orbit Type (Scatterplot)
- Flight Number by Launch Site (Scatterplot)
- Payload Mass by Launch Site (Scatterplot)
- Payload Mass by Flight Number (Scatterplot)

Thank you!